

KinaseCanvas: An Integrated Resource for Curated and Novel Kinase-Substrate Interactions

Chinmaya U. Joisa^{a,*}, Ian C. McCabe^{a,c}, Michael P. East^b, Xianlu L. Peng^{a,b},
Shawn M. Gomez^{b,d}, Jen Jen Yeh^{a,b,e}

^a*Lineberger Comprehensive Cancer Center, University of North Carolina at Chapel Hill, Chapel Hill, USA,*

^b*Department of Pharmacology, University of North Carolina at Chapel Hill, Chapel Hill, USA,*

^c*Department of Cell Biology and Physiology, University of North Carolina at Chapel Hill, Chapel Hill, USA,*

^d*Joint Department of Biomedical Engineering, University of North Carolina at Chapel Hill and North Carolina State University, Chapel Hill / Raleigh, USA,*

^e*Department of Surgery, University of North Carolina at Chapel Hill, Chapel Hill, USA,*

Text added or changed in this revision is shown in blue.

1. Abstract

Protein phosphorylation is a cornerstone of human physiology, regulating pathways critical for development, metabolism, and disease. While curated databases like PhosphoSitePlus and iPTMnet provide high-confidence kinase-substrate interactions (KSIs), they remain biased toward well-studied signaling nodes, leaving the vast majority of the “dark” kinome unconnected. Recent specificity atlases have begun to fill this gap by profiling motif specificities for nearly all human kinases, yet these raw predictions are often too dense and noisy for direct biological interpretation. Here, we present KinaseCanvas, an integrated resource that places these motif predictions within the context of high-confidence curated data. By applying stringent biological filters for subcellular co-localization and phosphosite functional importance, we distill **tens of millions** of raw predictions into **21,916** high-confidence novel interactions and integrate them with **8,364** curated **interactions**, yielding **30,280 distinct kinase-substrate pairs**. This integration **increases the share of interactions involving understudied kinases roughly 4.6-fold, from 3.4% to 15.7%**, and provides a context-aware framework for hypothesis generation, illuminating poorly characterized regions of the human kinome.

*Corresponding author

24 Database URL: <https://cujoisa.github.io/KinaseCanvas/>

25 **2. Introduction**

26 Protein phosphorylation is a pervasive regulatory mechanism that controls
27 virtually every major cellular process. Public repositories now catalogue well
28 over 200,000 human phosphosites distributed across the large majority of the
29 human proteome [1, 2], and modelling that accounts for the incomplete and
30 biased coverage of tryptic phosphoproteomics places the true totals higher
31 still [2, 3]. Estimates of how many of these sites are individually functional
32 are considerably lower [4, 5, 6], so the operative problem is not enumeration
33 but prioritization. Mapping which phosphorylation enzyme (kinase) modifies
34 which protein (substrate) is therefore essential for understanding signal trans-
35 duction, drug response and disease mechanisms, but this map remains far from
36 complete. Over the past two decades, substantial effort has been put into
37 building reference resources for kinase-substrate interactions (KSIs). Databases
38 such as PhosphoSitePlus, EPSD and iPTMnet curate experimentally observed
39 phosphosites and, where possible, annotate the upstream kinases based on low-
40 throughput biochemistry, targeted phosphoproteomics and manual literature
41 curation [1, 7, 8]. Additional resources including Phospho.ELM, PhosphoNet-
42 works and SIGNOR provide complementary KSI annotations derived from both
43 focused and high-throughput studies [9, 10, 11], although these differ substan-
44 tially in scope and currency: Phospho.ELM has not been updated since 2010,
45 and PhosphoNetworks reports kinase-substrate relationships measured on pro-
46 tein microarrays without resolving the modified residue [12, 10]. PhosphoNET
47 additionally provides sequence-based predictions of kinase-phosphosite pairings
48 for 486 human protein kinases, using scoring matrices that span the -7 to +7
49 positions around the phosphoacceptor, alongside curated sites [13]; the com-
50 panion KiNECTOR resource (<http://www.kinector.ca>) aggregates over 25,000
51 human kinase-substrate-phosphosite relationships into navigable pathway maps.
52 However, these curated datasets are heavily skewed towards a small subset of
53 well-studied kinases, substrates and pathways, leaving large regions of the human
54 kinome (including understudied “dark” kinases) sparsely connected or completely
55 unmapped [14].

56 In parallel, unbiased experimental and computational approaches have recently
57 produced near-comprehensive “specificity atlases” for the human kinome. Using
58 peptide-library based specificity profiling, Johnson et al. systematically deter-
59 mined optimal substrate motifs for most human serine/threonine kinases and
60 used these motifs to score reported human phosphosites [15]. More recently,
61 Yaron-Barir et al. generated an analogous atlas for the human tyrosine kinome
62 [16]. Together, these resources greatly expand the theoretical kinase-phosphosite
63 search space: for each phosphosite, nearly all known kinases receive a motif-based
64 compatibility score, yielding tens of millions of candidate KSIs. While these
65 atlases have already been shown to recover known signaling relationships and
66 highlight kinase dysregulation in disease contexts [15, 16], their full potential as a

67 systems-level KSI prior remains underexploited. Current visualization tools, such
 68 as the Kinase Library [15] and earlier motif scanners such as Scansite [17], focus
 69 on site-level scoring and lack interactive network visualization of kinase-substrate
 70 relationships. Furthermore, while methods like Kinase-Substrate Enrichment
 71 Analysis (KSEA) are widely used to infer kinase activity, they typically rely on
 72 curated databases or require manual integration of specificity matrices.

73 Moreover, profiling thousands of phosphosites against hundreds of kinases
 74 generates an enormous combinatorial prediction space: most sites have multiple
 75 plausible motif matches, and naïve motif scanning across the phosphoproteome
 76 leads to high false-positive rates and spurious kinase-substrate couplings [18, 19].
 77 A major challenge is therefore that motif-based scores alone do not ensure that
 78 a kinase-substrate pair is physically or functionally plausible. Kinases and
 79 substrate pairs must co-localize in the cell, be expressed in the same tissues or
 80 conditions, and the modified site itself must be known or predicted to have some
 81 functional consequence. To address this, recent studies have demonstrated the
 82 value of integrating orthogonal data layers. In particular, Ochoa et al. developed
 83 a machine-learning approach that aggregates features ranging from evolution-
 84 ary conservation and structural constraints to regulation across thousands of
 85 phosphoproteomics experiments. Their functional score effectively discriminates
 86 regulatory sites from non-functional ‘noise,’ substantially improving the bio-
 87 logical interpretability of phosphoproteomic datasets [5, 20]. When combined
 88 with orthogonal constraints such as subcellular localization, these functional
 89 indicators could provide a powerful filter for identifying physiologically relevant
 90 kinase-substrate relationships.

91 Here, we build on these advances to construct an integrated resource of hu-
 92 man kinase-substrate relationships that combines literature-curated KSIs with
 93 motif-based predictions from the serine/threonine and tyrosine kinome specificity
 94 atlases. We apply a set of simple but stringent biological filters based on (i)
 95 motif rank, (ii) subcellular co-localization using the Human Protein Atlas (HPA),
 96 and (iii) a data-driven phosphosite functional score [5, 21, 15, 22]. An overview
 97 of the data integration and filtering strategy is shown in Figure 1(A), includ-
 98 ing an example TRAF2 and NCK-interacting protein kinase (TNIK)-centred
 99 subnetwork. The resulting network is accessible through the KinaseCanvas Ex-
 100 plorer, an interactive web application that allows users to query specific kinases
 101 or substrates, visualize their local interaction neighbourhoods, and seamlessly
 102 filter connections based on curation status or functional evidence. This unified
 103 framework enables researchers to distinguish novel from previously reported
 104 interactions and to systematically assess coverage of both well-studied and dark
 105 kinases.

106 Alt text: Workflow diagram showing integration of curated Kinase-Substrate
 107 sets with motif predictions, co-localization filters, and functional scores.

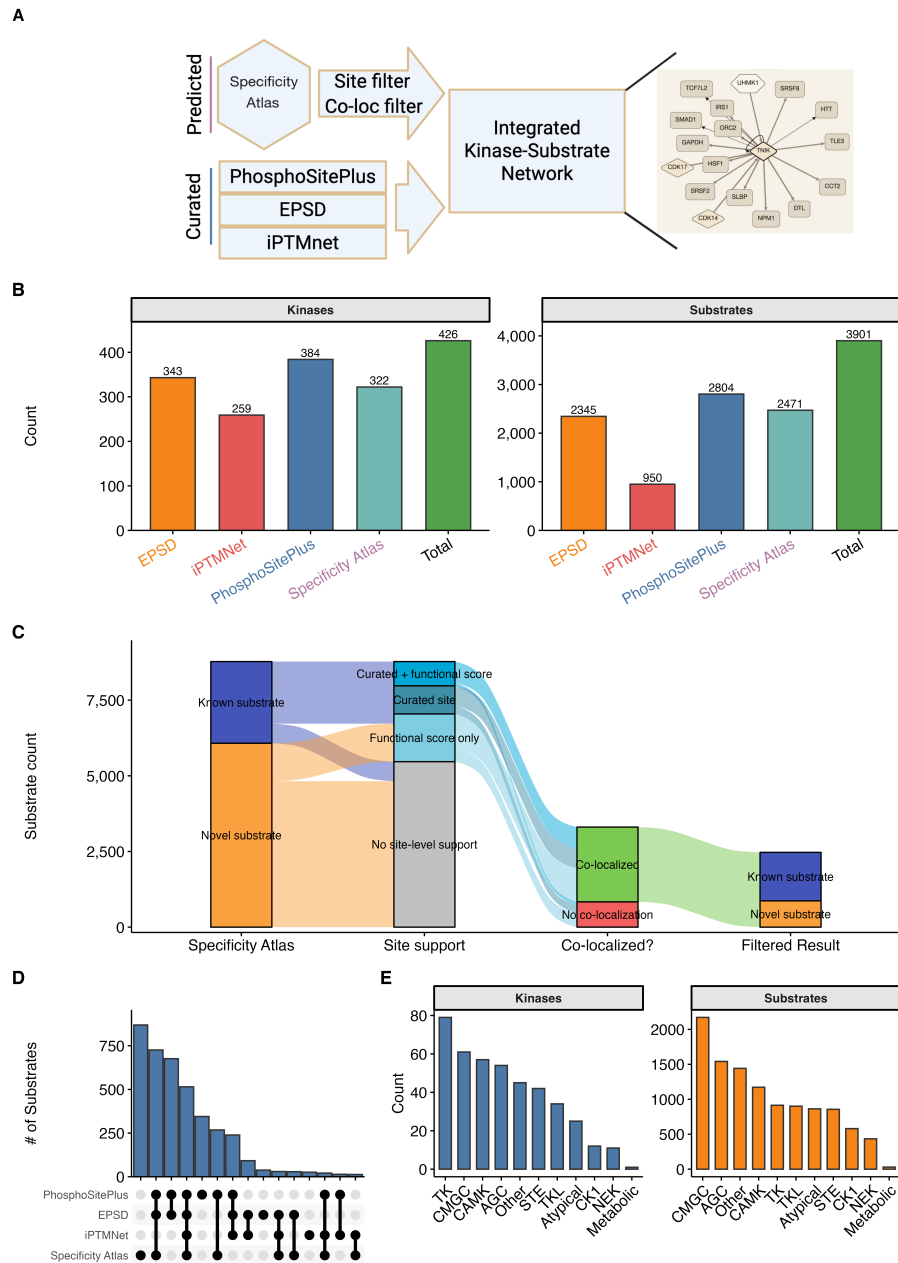


Figure 1: Construction of the integrated kinase-substrate network. A) Overview of the integration strategy. Motif-based predictions from the specificity atlas are filtered by site-level and co-localization criteria and merged with curated KSIs from PhosphoSitePlus, EPSP and iPTMnet to generate an integrated kinase-substrate network. An example [TNIK-centred neighbourhood from the KinaseCanvas Explorer](#) illustrates how curated (dark) and predicted (light) interactions are combined. B) Numbers of kinases (left) and substrates (right) contributed by each source and in the integrated total. C) Alluvial diagram tracking substrates through successive filtering stages: known vs. novel status, site-level support, and co-localization, to the final high-confidence set. The diagram is conditioned on substrates already retained by the initial motif and promiscuity filters and is not a sample of the human phosphoproteome. D) UpSet plot showing overlap of substrates across the four main KSI sources, highlighting both unique and shared substrate sets. E) Distribution of kinases (left) and substrates (right) across major kinase families in the integrated network.

3. Materials and Methods

3.1. Curated kinase-substrate interactions

We first assembled a non-redundant reference set of human kinase-substrate-site relationships (“curated KSIs”) from multiple databases. Primary sources included PhosphoSitePlus (PSP), the Eukaryotic Phosphorylation Site Database (EPSD) and iPTMnet, which catalogue experimentally observed phosphosites and often annotate upstream kinases based on low-throughput experiments or kinase assays [23, 7, 8].

As a starting point, we downloaded the pre-assembled curated KSI table from the KiNet web portal, which aggregates KSIs from PhosphoSitePlus, EPSD and iPTMnet and provides kinase annotations [24]. We then re-processed this table to harmonize identifiers and used available site-level functional annotations needed for our analyses. Briefly, we extracted the kinase gene symbol, substrate protein, phosphosite (modified residue and position), source database and/or PMID. Gene and protein identifiers were harmonized by mapping to official HGNC gene symbols and UniProt accessions (current versions as of January 10, 2026). When multiple isoforms existed, phosphosites were mapped to the reviewed (Swiss-Prot) sequence based on the human proteome when possible. Phosphosite notation was standardized to a single-letter residue code plus position (e.g. “S123”). Redundant entries arising from overlap across databases were collapsed, while preserving provenance information (all contributing databases and PMIDs).

Across PhosphoSitePlus, EPSD, iPTMnet and the specificity atlas we ultimately obtained 13,448 unique phosphosites, with PhosphoSitePlus contributing the largest number (9,339 sites) and EPSD (7,525 sites) and the specificity atlas (5,308 sites) contributing substantial additional coverage (Supp. Figure 1(A)). The overlap structure of these site sets, including sites unique to individual resources and those shared across multiple databases, is summarized in Supp. Figure 1(B). The resulting curated KSI set serves as a reference baseline for all subsequent analyses, and coverage of kinases and substrates contributed by each source is shown in Figure 1(B).

3.2. Kinome-wide motif-based substrate predictions

To substantially expand beyond known KSIs, we integrated motif-based predictions from the human kinome specificity atlases for serine/threonine and tyrosine kinases. Johnson et al. used position-specific scoring matrices derived from peptide library screens to define optimal substrate motifs for the vast majority of human serine/threonine kinases and subsequently scored reported human phosphosites against each kinase-specific motif [15]. These matrix-based approaches have been complemented by more recent deep-learning architectures for phosphorylation site prediction [25]. More recently, Yaron-Barir et al. generated an analogous atlas for the human tyrosine kinome [16]. We and others collectively refer to these resources as the “specificity atlas”.

We obtained the per-kinase, per-phosphosite prediction scores from the supplementary materials. For each phosphosite (UniProt accession + position),

the atlases provide a quantitative score or percentile rank reflecting how well the local sequence matches the optimal motif for each kinase [15, 16]. The complete, unfiltered specificity dataset—containing motif ranks, percentiles, and promiscuity indices for nearly 28 million kinase-substrate-site combinations—is provided as a supplemental resource (Supp. Table 1). From this prediction matrix, we extracted high-confidence motif-based KSIs using a percentile threshold. Following the original studies, we focused on phosphosite-kinase pairs in which the site ranked within the top 10% of all phosphosites scored for that kinase (≥ 90 th percentile) [15, 16]. This conservative cutoff retains sites that are strongly preferred by the kinase’s motif and reduces the inclusion of weak, low-specificity matches. Because a single stringent tier structurally excludes kinases whose specificity is dominated by docking interactions rather than by the catalytic cleft (see Discussion), we additionally release a relaxed tier comprising kinase-site pairs between the 75th and 90th motif percentiles that satisfy the same contextual criteria. This layer is released as a separate, labelled table (see Data Availability) and is not part of the high-confidence set.

To further mitigate the problem of phosphosites that match the motifs of many kinases, we utilized the “promiscuity index” provided for each scored phosphosite, defined as the number of kinases for which that site ranks at or above the 90th percentile. Sites at or above the median promiscuity index (21 kinases) were excluded from the motif-based prediction set, as a site compatible with many kinases’ motifs carries little information about which of them acts on it. The remaining set of motif-supported KSIs was then passed to contextual filters described below.

3.3. Contextual filtering

3.3.1. Subcellular co-localization

Since kinase activity is constrained by spatiotemporal co-localization with substrates, we filtered motif-based KSIs based on subcellular localization. We used the HPA subcellular annotations, which provide antibody- and imaging-based localization calls for thousands of human proteins across 30 subcellular structures [22] (HPA subcellular location dataset, downloaded 10 January 2026). The mapping of gene-level subcellular compartmentalization used in this study is provided as a supplemental table (Supp. Table 2). For each kinase and substrate, we retrieved all reported compartments annotated at the “Enhanced”, “Supported” or “Approved” reliability levels, excluding “Uncertain” calls. For every predicted kinase-substrate pair, we then computed the intersection of kinase and substrate localization sets and recorded the number of shared compartments, which is released as a continuous column rather than only as a binary flag. Pairs with at least one shared compartment were considered co-localized and retained; pairs with no compartmental overlap were flagged as unlikely to occur in vivo and removed from the high-confidence set. Proteins absent from HPA necessarily have no shared compartment and are therefore treated as non-co-localized, so missing annotation and genuine spatial separation are not distinguished by this filter. The distribution of the fraction of motif-predicted substrates that

co-localize with each kinase is shown in Figure 2(A); the median kinase retains 37.2% of its predicted substrates after applying this co-localization filter. We note that HPA localizations are neither exhaustive nor condition-specific, so some genuine context-dependent interactions may be lost. However, requiring at least one documented shared localization substantially increases the plausibility of retained KSIs.

3.3.2. Phosphosite functional score

To prioritize phosphosites with higher likelihood of regulatory impact, we integrated the phosphosite functional score developed by Ochoa et al., who re-analysed thousands of phosphoproteomics experiments to build a reference human phosphoproteome and then trained a gradient boosting model on sites of known function to integrate 59 features, grouped as mass-spectrometry evidence, phosphosite regulation, structural environment and evolutionary conservation, into a single score of functional relevance [5]. The resulting score ranges from 0 to 1, with higher values indicating a greater probability that a site is functionally important.

We downloaded the pre-computed human phosphosite functional scores and mapped them to our phosphosite identifiers (provided as Supp. Table 3). For each predicted KSI, we annotated the substrate site with its functional score and designated a site as “likely functional” if its score was ≥ 0.5 . This is the threshold recommended by the authors of the score, who report that at a 0.5 cut-off approximately 10% of the phosphoproteome is retained while 50% of true-positive functional sites are recovered, making it a useful cut-off for initial prioritization [5]; comparable thresholds have been applied in subsequent work [16]. Because any fixed threshold is arbitrary at its boundary, we additionally report a sensitivity analysis across the range 0.3-0.7 and release a continuous confidence score (see below). As expected, tyrosine sites were markedly enriched for site-level support compared to serine or threonine sites (36.1% vs. 12.8% and 15.0% of unique sites, respectively; Figure 2(B)). Here, as in Figure 2(B), a site counts as supported if it is either already curated or carries a functional score above the cutoff; restricting to functional-score support alone among sites that carry a score gives 26.5%, 7.7% and 8.6%. The difference between residue classes is highly significant ($\chi^2 = 1848.1$, $df = 2$, $p < 2 \times 10^{-16}$; $n = 5,132$ tyrosine, 34,754 serine and 7,844 threonine sites). When stratified by source database, the proportion of scored sites classified as likely functional ranges from 45.4% (specificity atlas layer) and 47.4% (PhosphoSitePlus) through 50.1% (EPSD) to 62.2% (iPTMnet) (Figure 2(C) and Supp. Figure 1(C)). The atlas-derived layer is not enriched relative to the curated sources because, under the disjunctive site-level criterion described below, a site can enter it on the strength of curation alone; the comparison is therefore most informative for the curated sources, which are not filtered on this score. Unless otherwise noted, analyses focusing on functional sites consider only motif-based KSIs whose substrate sites passed this functional score threshold.

Alt text: Statistical distribution of phosphosite functional scores across different

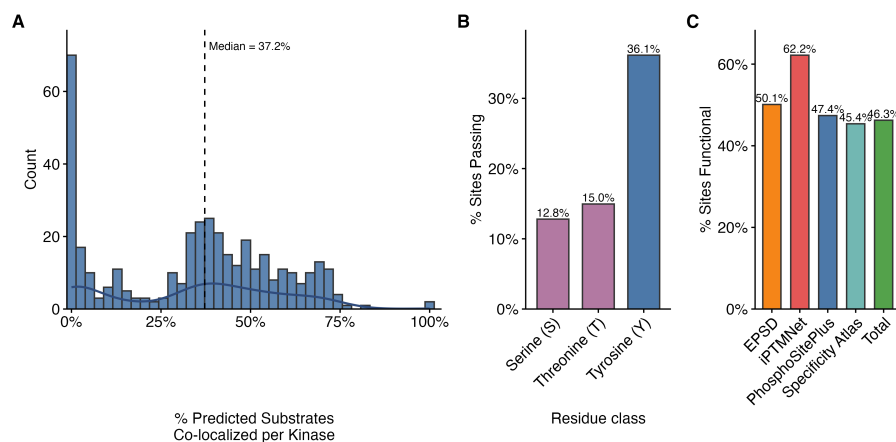


Figure 2: **Specificity Atlas Filtering Results.** A) Distribution of subcellular co-localization percentages for predicted kinase-substrate pairs per kinase. Co-localization was determined using HPA subcellular location data, requiring at least one shared compartment between the kinase and substrate; percentages are computed over each kinase's predicted substrates, counting those without HPA annotation as non-co-localized, as the filter does. The dashed line indicates the median co-localization (37.2%). B) Percentage of predicted sites passing filtering criteria across residue classes (Serine, Threonine, and Tyrosine). "Pass" is defined as a site being either previously curated in primary databases (PhosphoSitePlus, EPSPD, iPTMNet) or possessing a high functional score (Ochoa score ≥ 0.5). C) Percentage of scored sites classified as likely functional (functional score ≥ 0.5) in each data source.

239 residue types (Y, S, T) and source databases.(A) shows co-localization fraction
240 by kinase;(B) shows functional score enrichment for tyrosine sites.

241 3.4. Integration of motif, localization and function

242 Combining the motif, promiscuity, localization and functional filters, we defined
243 a high-confidence predicted KSI set as those kinase-substrate-site triplets that
244 satisfy all of:

- 245 1. Motif support: the substrate site ranks in the top 10% of all phosphosites
246 scored for that kinase (≥ 90 th percentile) in the relevant serine/threonine
247 or tyrosine atlas.
- 248 2. Specificity: the site falls below the median promiscuity index, excluding
249 sites that are top-ranked by many kinases.
- 250 3. Co-localization: kinase and substrate share at least one HPA-annotated
251 subcellular compartment.
- 252 4. Site-level support: the substrate site either has a phosphosite functional
253 score ≥ 0.5 or is already curated in PhosphoSitePlus, EPSD or iPTMnet.

254 Criterion 4 is deliberately a disjunction rather than a conjunction. A site that
255 has already been curated from the primary literature carries direct experimental
256 evidence of relevance, so requiring it to additionally clear a machine-learned
257 functional-score threshold would discard well-established sites on the basis of
258 a predictor. Curated and functional-score support are reported as separate
259 columns so that either can be required independently.

260 The effect of these filters on the substrate set is visualized as an alluvial
261 diagram in Figure 1(C), with per-stage substrate counts given in Supp. Table 4.
262 It tracks substrates from known vs. novel status, through site-level support and co-
263 localization, to the final set. Although some true interactions will undoubtedly
264 be excluded (e.g. transient co-localization not captured by HPA, functional
265 sites below the threshold, or kinases with context-dependent specificity), we
266 deliberately favour specificity over sensitivity to generate a conservative, high-
267 confidence prediction set. Each predicted KSI is also cross-referenced to the
268 curated KSI list; interactions present in curated databases are flagged as “known”,
269 whereas motif-only interactions with no curated evidence are classified as novel
270 predictions contributed by this resource.

271 3.5. Species and isoform handling

272 The resource is restricted to human proteins (NCBI taxonomy 9606). Every
273 protein in the curated KiNet table is a reviewed (Swiss-Prot) human UniProt
274 entry, and we performed no cross-species orthologue mapping of our own, so a
275 site enters the resource only if a contributing database reports it on the human
276 protein. All phosphosite positions refer to the canonical UniProt sequence of the
277 substrate: neither the curated table nor the specificity atlases use isoform-specific
278 accessions. Every site is checked against that canonical sequence (see below),
279 and the UniProt accession is released alongside every site so that positions can
280 be verified independently.

281 3.6. Substrate and kinase identity resolution

282 Resolving protein identity from gene symbols alone is unreliable for this appli-
283 cation, because many legacy protein names are aliases of several distinct genes.
284 We therefore resolve substrate identity from the UniProt accession supplied
285 by the specificity atlases, falling back to the gene symbol only where it maps
286 unambiguously to a single gene. Kinase identity is resolved from the atlas column
287 names (which use Manning/CORAL-style short names such as P38A, CK1D and
288 MEK1 rather than HGNC symbols) against a curated kinome reference table,
289 then against unambiguous aliases, with a small explicit override list for names
290 that neither source covers; the pipeline halts if any atlas kinase name remains
291 unresolved rather than silently discarding it. Every released kinase-substrate-site
292 triplet is checked automatically against the source atlas to confirm that the site
293 belongs to the accession it is reported under. Every site is additionally checked
294 against its canonical UniProt sequence, which also yields the -7/+7 flanking
295 sequence shown for each site (verified for 95.4% of sites; unverified sites are left
296 without one rather than shown an unverified guess).

297 3.7. Confidence scoring

298 Alongside the binary criteria above, each predicted interaction carries a con-
299 tinuous confidence score. Motif percentile and functional score are each passed
300 through a monotonic, continuously differentiable logistic transform centred on
301 the corresponding threshold, and localization evidence is expressed as a saturat-
302 ing function of the number of shared HPA compartments rather than a binary
303 flag. The composite score is the unweighted mean of the three components, and
304 all three components are released as separate columns so that users can reweight
305 or threshold them independently. This is a transparent re-weighting rather than
306 a calibrated probability: a fully probabilistic treatment would require labelled
307 negative examples, which do not exist at kinome scale [26, 27].

308 3.8. Information content of the evidence layers

309 To quantify how much each evidence layer contributes and how far the layers
310 overlap, we recorded the full joint contingency table of the three indicators
311 (co-localization, curated-site status and functional-score support) over all motif-
312 passing kinase-site pairs, before any of them had been applied. From this table we
313 computed marginal Shannon entropies, pairwise mutual information, normalised
314 redundancy (mutual information divided by the smaller of the two marginal
315 entropies) and conditional mutual information, together with a leave-one-filter-
316 out analysis quantifying how many pairs each layer uniquely removes.

317 3.9. Annotation of dark kinases

318 To assess how the integrated network covers dark kinases, we used the DarkKi-
319 naseTools R package from the NIH Illuminating the Druggable Genome (IDG)
320 programme to classify kinases as “understudied (dark)” or “well-studied (light)”
321 based on publication counts, functional annotations and available reagents [14].

322 This classification was mapped onto our KSI network, enabling summary statis-
 323 tics on the proportion of curated vs. predicted substrates assigned to dark kinases.
 324 We also summarised the fraction of substrates that are novel (predicted only) for
 325 each kinase family (Figure 3(A)) and quantified the fraction of kinase–substrate
 326 pairs involving dark kinases in the curated set versus the curated-plus-specificity
 327 atlas integrated network (Figure 3(B)).

328 3.10. Implementation

329 All data integration, filtering and visualisation were performed in R using the
 330 tidyverse ecosystem for data manipulation and plotting. Kinome annotations and
 331 dark/light classifications were obtained via DarkKinaseTools and associated IDG
 332 resources [14]. Upset plots showing overlap between resources (for phosphosites
 333 and substrates) were generated using the [ggupset package](#) (Supp. Figure 1(B);
 334 Figure 1(D)). Scripts used to download source datasets, perform identifier
 335 harmonisation, apply filters and generate figures are provided in GitHub to
 336 facilitate reproducibility and reuse.

337 4. Results

338 4.1. Structure and coverage of the integrated resource

339 The final resource is organised into two primary tables:

- 340 1. **Kinase-substrate pair table.** This table lists each unique kinase–
 341 substrate pair (gene symbols), with columns indicating:
 - 342 • whether the pair is supported by curated databases, motif-based
 343 predictions, or both;
 - 344 • the number, identity and directionality of phosphosites linking the
 345 kinase and substrate;
 - 346 • source databases and PMIDs for curated evidence;
 - 347 • motif-related annotations (serine/threonine vs. tyrosine atlas);
 - 348 • contextual evidence flags (co-localization, functional score above
 349 threshold); and
 - 350 • dark/light status of the kinase.
- 351 2. **Phosphosite table.** This table lists each unique substrate site (UniProt
 352 accession, residue, position) together with its assigned kinases and annota-
 353 tions, including:
 - 354 • motif-based scores and ranks for all compatible kinases;
 - 355 • flags for high-confidence motif assignments (top 10%);
 - 356 • phosphosite functional score (continuous) and functional classification
 357 (≥ 0.5 vs. < 0.5);
 - 358 • curated vs. predicted [status](#);
 - 359 • [the -7/+7 flanking sequence, verified against the canonical sequence;](#)
 360 [and](#)
 - 361 • basic protein metadata.

Across all sources, the integrated resource covers 426 kinases and 3,901 substrate proteins (Figure 1(B)). PhosphoSitePlus and EPSD contribute the largest numbers of substrates, while the specificity atlas brings in additional substrates not present in curated databases (Figure 1(B)). At the site level, we catalogue 13,448 unique phosphosites across PhosphoSitePlus, EPSD, iPTMnet and the specificity atlas (Supp. Figure 1(A)), with substantial but incomplete overlap between resources (Supp. Figure 1(B)).

In terms of kinase-substrate pairs, the curated KSI set contains 8,364 distinct pairs, while the high-confidence motif-based predictions add 21,916 novel pairs not present in curated databases, for a total of 30,280 distinct pairs (Figure 3(C)). Most kinases gain dozens of novel substrates: the distribution of novel substrates per kinase is right-skewed, with a median of 47 novel substrates among kinases that also have curated substrates, and some kinases gaining more than 350 (Figure 3(D)).

Alt text: Comparison of kinase coverage and substrate counts between curated databases and the integrated resource. Shows a 3.6-fold increase in KSI pairs and highlights expansion for dark kinases across different families.

4.2. Functional and family-level properties

Site-level support is most frequent for tyrosine sites and for sites drawn from iPTMnet (Figure 2(B) and Supp. Figure 1(C)). Of the 3,307 substrates with site-level support, 2,471 (75%) have at least one kinase-site pair that is also co-localized and are retained (Figure 1(C); Supp. Table 4).

Kinase and substrate coverage varies across kinase families (Figure 1(E)). TK and CMGC kinases are most numerous, whereas AGC, CAMK and “other” families contribute comparable numbers of kinases but differ in substrate counts (Figure 1(E)). When focusing on novel substrates, dark kinase families show particularly high fractions of predicted-only substrates: “other” and atypical kinases have 66.6% and 62.0% of their substrates classified as novel, respectively, while CAMK and CMGC kinases have 50.7% and 37.5% (Figure 3(A)). Incorporating the specificity atlas increases the fraction of kinase-substrate pairs involving dark kinases roughly 4.6-fold, from 3.4% in curated databases alone to 15.7% in the integrated network (Figure 3(B)), underscoring the value of motif-based predictions for illuminating poorly studied regions of the kinome.

4.3. Robustness of the filtering criteria

Because both the motif and functional-score criteria are applied as thresholds, we asked how sensitive the resource is to where those thresholds are placed. Sweeping the functional-score cut-off from 0.3 to 0.7 and the motif percentile from the 80th to the 99th (Supp. Figure 2) changes the size of the high-confidence set substantially – at the 90th motif percentile, the number of retained pairs falls from 58,319 to 13,815 across the functional-score range – but leaves the qualitative conclusions intact. The proportion of retained predicted pairs involving a dark kinase varies only between 19.1% and 20.8% across that same functional-score

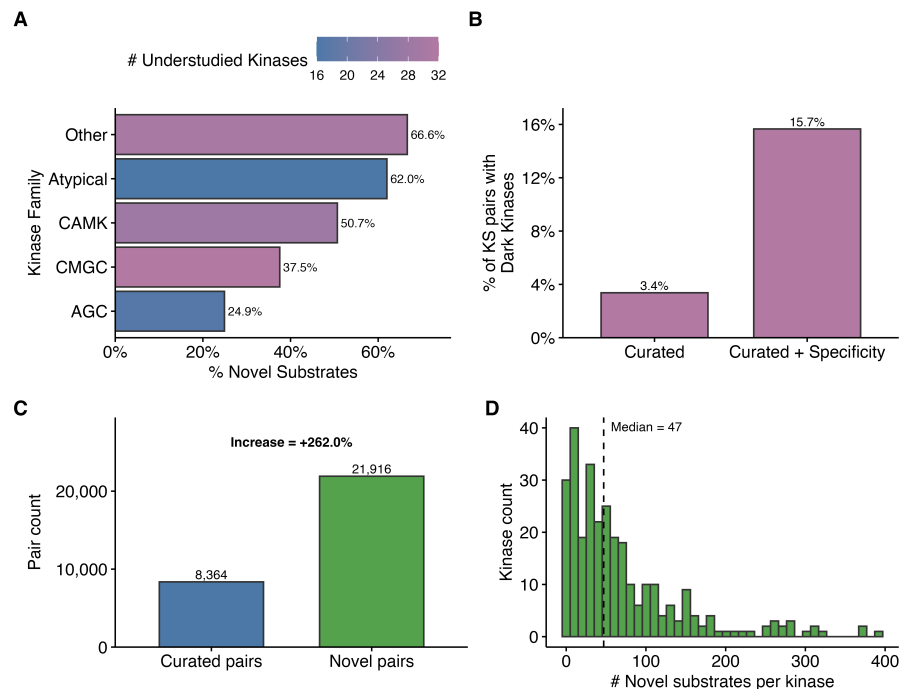


Figure 3: Expansion of the known kinase-substrate network. A) Fraction of novel substrates across major kinase families. Bars are colored by the number of dark kinases within each family, as defined by the IDG (Illuminating the Druggable Genome) classification. B) Impact of the specificity atlas on the representation of dark kinases. The bar plot shows the percentage of KS pairs involving “Dark” kinases in curated databases versus the combined “Curated + specificity atlas” dataset, demonstrating a **substantial** increase in the inclusion of dark signaling components. C) Global expansion of the kinase-substrate (KS) network. The bar plot compares the total number of curated pairs against novel pairs introduced by the specificity atlas, showing a **3.6-fold** increase in network **coverage** (+262.0%). D) Histogram showing the distribution of novel substrates per kinase for kinases overlapping between curated sources and the specificity atlas. The dashed line indicates a median of **47** novel substrates per kinase.

range, and the number of kinases represented varies between 318 and 325. The sweep re-implements the filters independently; at the operating point used throughout (90th percentile, functional score 0.5) it retains 22,804 predicted pairs, against 22,797 in the released resource.

We also asked how much information each evidence layer actually contributes, and how far the layers overlap. Across all motif-passing kinase-site pairs, subcellular co-localization carries the highest marginal entropy (0.896 bits, versus 0.418 bits for curated-site status and 0.430 bits for functional-score support). Pairwise mutual information between the three layers is very small: 0.0008 bits between co-localization and curated status, 0.0007 bits between co-localization and functional-score support, and 0.0082 bits between curated status and functional-score support, giving normalised redundancies of 0.19%, 0.16% and 1.95% respectively, and conditioning on the third indicator does not change this. The layers are therefore close to orthogonal, and each is doing distinct work: in a leave-one-out analysis, co-localization uniquely removes 47,029 kinase-site pairs that satisfy every other criterion, and omitting it would inflate the high-confidence set by 182%. This filter is not trivially satisfied: the median gene has HPA annotations for two compartments and only 2.6% of annotated genes are assigned to five or more.

The specificity atlases score only phosphosites that have already been observed in phosphoproteomics experiments, and because that literature is dominated by immortalized and cancer cell lines, we expected coverage to be uneven with respect to tissue-restricted proteins. It is. Classifying substrates by HPA RNA tissue specificity, 70.4% of substrates with low tissue specificity gain at least one novel predicted interaction, falling monotonically to 57.8% for tissue-enhanced, 46.9% for group-enriched and 40.6% for tissue-enriched proteins. Conversely, the proportion of substrates for which we contribute nothing beyond existing curation rises from 29.6% to 59.4% across the same gradient. Predictions are therefore systematically sparser for specialized proteins that define differentiated cell types, and users working on tissue-restricted biology should expect the resource to be less complete for their proteins of interest.

4.4. Interactive exploration and visualization

To make these integrated data accessible to the broader community, we developed KinaseCanvas Explorer, an interactive web application that enables users to query, filter, and visualize the kinase-substrate network. The explorer provides a unified interface where users can seed the graph with a kinase or substrate of interest to generate a local interaction neighbourhood (see interface in Figure 4(A, B) and a representative network in Figure 4(C)). Key features include:

- **Kinase- or substrate-first querying:** The graph can be seeded from either a kinase or a substrate, with autocomplete on both fields. Seeding with a substrate returns all upstream kinases, curated and predicted, so

the resource can be used to ask which kinases might act on a protein of interest as readily as the reverse.

- **Site-level interrogation of a specific pair:** A pair mode restricts the view to a single kinase-substrate link and ranks the candidate sites on that substrate. This addresses the common experimental situation in which a kinase is known to phosphorylate a protein (e.g., from a band shift on a Phos-tag gel or a pan-phosphotyrosine blot) but the modified residue is unidentified.
- **Site sequence context:** Each site's -7/+7 flanking sequence is shown with the phosphoacceptor highlighted, verified against the canonical UniProt sequence for every site in the resource.
- **Quantitative evidence on every interaction:** The motif percentile, the kinase's rank for that site among all kinases, the phosphosite functional score, the number of shared subcellular compartments and the composite confidence score are all displayed in a filterable table of the interactions in view and carried into every table export. A minimum-confidence slider filters predicted edges on the composite score.
- **Explicit evidence provenance:** Curated-site status and functional-score support are reported as separate fields, so a site known to be functional from the primary literature is distinguishable from one predicted to be functional, and the view can be restricted to curated pairs or to functional sites.
- **Visual encoding:** Node shape and colour encode role (kinase, substrate, or both), and for predicted edges the line thickness and opacity scale with the composite confidence score, so relative support is visible and not merely filterable.
- **Legible neighbourhoods for promiscuous kinases:** Highly connected kinases would otherwise render as unreadable hairballs; the most connected, CDK1, has 629 substrates. A seeded neighbourhood is therefore capped at 75 interactions by default, chosen under a stated ordering (curated first, then by descending composite confidence, ties broken alphabetically), with the count shown and the cap removable in one click. The cap does not affect the 63% of kinases, and 96% of all proteins in the resource, whose neighbourhoods contain 75 or fewer interactions.
- **Click-to-inspect detail panel:** Selecting a protein shows its role, UniProt link, numbers of substrates and upstream kinases, and its highest-ranked interactions. Selecting an interaction shows its evidence, sources and PMIDs, together with a per-site table giving each site's flanking sequence, curation status, functional score, motif percentile, kinase rank and shared compartments, with curated sites listed first and the rest ranked by motif percentile.
- **Display options:** Five layout algorithms (fCoSE, CoSE, CoSE-Bilkent, Concentric and Breadthfirst) are available, together with adjustable label size and optional edge labels and arrows.
- **Export options:** Network images can be exported as PNG or PDF, and both edge-level and site-level tables can be downloaded as CSV for either

the current selection or the complete network.

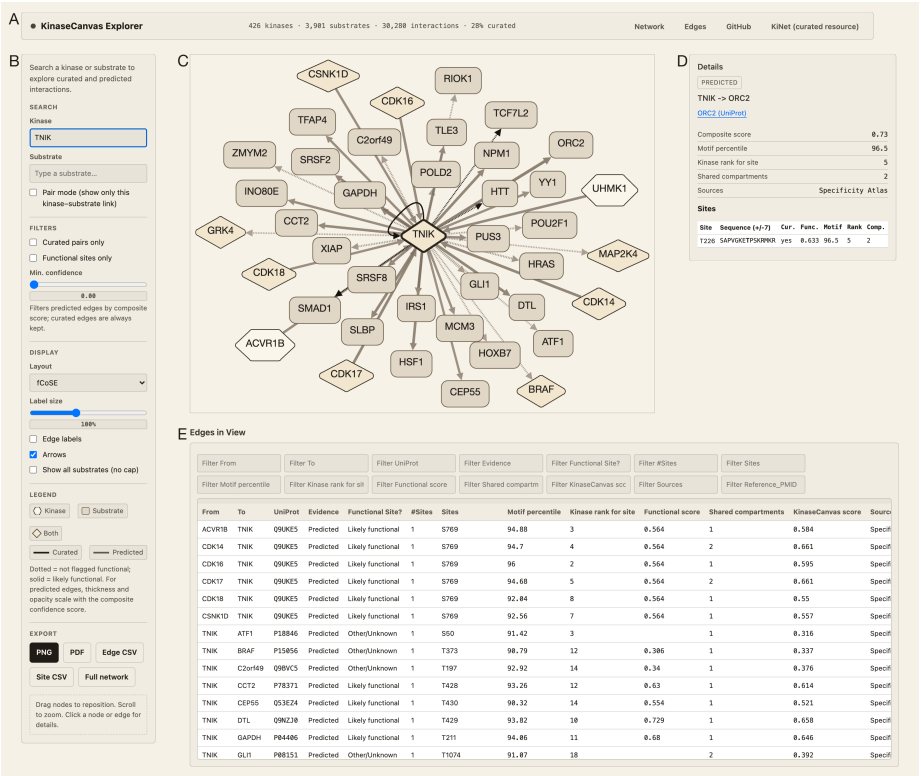


Figure 4: **The KinaseCanvas Explorer.** A) Header bar, reporting summary statistics for the released resource. B) Sidebar: kinase- and substrate-first search with autocomplete and a pair mode, filters for curated pairs, functional sites and minimum composite confidence, display options, legend and export buttons. C) Network view seeded with TNK (TRAF2 and NCK-interacting kinase). Node shape and colour encode role (hexagon, kinase; rectangle, substrate; diamond, both); edge colour distinguishes curated (dark) from predicted (light) interactions; solid lines mark sites with functional support and dotted lines sites without; and the thickness and opacity of predicted edges scale with the composite confidence score. D) Detail panel for a selected interaction (TNK to ORC2), giving the evidence behind it and a per-site table with the flanking sequence, curation status, functional score, motif percentile, kinase rank and number of shared compartments. E) Filterable table of the interactions currently in view, which can be exported as CSV together with the site-level table.

Alt text: The KinaseCanvas Explorer interface, showing the header statistics, the search and filter sidebar, a TNK-seeded network, the detail panel for a selected interaction with its per-site evidence, and the filterable table of interactions in view.

5. Discussion

We have developed a comprehensive resource that combines literature-curated phosphorylation data with empirically measured, kinome-wide substrate specificity profiles to chart the human kinase–substrate network. This addresses a major gap in cell-signalling knowledge: despite decades of study, substrates are still unknown or poorly defined for most human kinases [9, 10, 14]. By leveraging recent motif profiles for ~90% of the human kinome [15, 16], we systematically expand substrate lists for hundreds of kinases and then constrain these predictions with biologically motivated filters for co-localization and phosphosite functionality. The result is a collection of KSI hypotheses that is both broad—increasing the number of candidate KSIs 3.6-fold relative to curated databases alone (Figure 3(C))—and enriched for plausible, potentially regulatory interactions (Figure 1(A); Figure 2(A); Supp. Figure 1(A)).

5.1. Relationship to existing resources

Traditional resources such as PhosphoSitePlus, EPSD and iPTMnet grow as new experimental data and curated literature are added [23, 7, 8]. These databases provide high-confidence KSIs, but their coverage is limited by experimental throughput and publication bias towards a small set of kinases and pathways [14]. The recently introduced KiNet web portal builds directly on these curated datasets by aggregating interactions from PhosphoSitePlus, EPSD and iPTMnet, annotating kinases with group membership and pathway/domain context, and providing interactive network visualisations for user-selected protein sets [24].

Our work is complementary to, and explicitly builds on, KiNet. We use the KiNet-integrated KSI network as the starting curated set, but then extend it in three major ways. First, we overlay kinome-wide, empirically measured substrate specificity atlases for serine/threonine and tyrosine kinases [15, 16] to infer motif-supported kinase–phosphosite edges that are not yet observed experimentally (Figure 3(C)). Second, we integrate orthogonal contextual information: subcellular localization from HPA [22] and phosphosite functional scores from Ochoa et al. [5] to filter and prioritize predicted interactions (Figure 1(A); Figure 2(A); Supp. Figure 1(A)). Third, we systematically quantify how these additions expand coverage for dark kinases and for different kinase families (Figure 3 (A, B)). In practical terms, whereas KiNet focuses on exploring known KSIs across pathways and protein sets, our resource aims to “fill in” missing edges by providing a ranked, context-aware set of candidate interactions centred on phosphosite-level properties.

The Kinase Library web portal [15] serves as the primary interface for the Johnson et al. and Yaron-Barir et al. atlases, allowing users to score specific peptide sequences against kinase motifs. However, it is designed for sequence-level queries rather than network exploration. Similarly, tools like the KSEA App enable kinase-substrate enrichment analysis but often rely on curated databases or require users to upload custom specificity matrices, and do not provide a browsable network of high-confidence predictions.

Earlier motif-based tools such as NetPhorest, Scansite and GPS used consensus sequence motifs and position-specific scoring matrices compiled from smaller sets of peptide library experiments or literature-derived substrates [18, 17, 28]. These resources proved useful for hypothesis generation but were limited in kinase coverage and often relied on consensus motifs shared across related kinases. The specificity atlases used here are broader in scope, profiling hundreds of human kinases with oriented peptide libraries and scoring essentially the entire known human phosphoproteome [15, 16]. *Intersecting* these empirically derived motif scores with orthogonal features *is intended to* improve the signal-to-noise ratio relative to motif scores alone. This multi-layer integration is aligned with a broader trend in systems biology toward combining complementary datasets and prior knowledge to infer more accurate signalling networks [5, 16, 11].

5.2. *Motif models derived from peptide libraries and from in vitro phosphoproteomics*

The motif layer of this resource is built on positional-scanning peptide library data [15, 16], which has inherent limitations. Peptide libraries measure the intrinsic preference of the catalytic cleft against a degenerate synthetic background, which gives uniform coverage across the kinome but can have modest signal-to-noise for individual kinases and, by construction, reports only what the active site sees. An orthogonal strategy is to phosphorylate dephosphorylated proteome extracts with individual recombinant kinases and read out the products by mass spectrometry, as done by Sugiyama et al. for 385 kinases [29]. This design presents kinases with protein substrates in native sequence context, and several groups have since converted the resulting data into position-specific matrices suitable for scoring. Poll et al. derived position-dependent Shannon information matrices for 384 kinases [30], and Invergo et al. built position-specific scoring matrices and naive Bayes classifiers from the same source [26].

These two experimental designs are complementary rather than competing. Poll et al. compared their mass-spectrometry-derived “logos” directly against the peptide-library motifs used here and reported a high degree of concordance, concluding that the best predictions are likely to come from combining both data types [30]. A systematic head-to-head comparison of peptide-library and in vitro phosphoproteomic specificity models is an important future direction as an independent benchmarking enterprise. In the meantime, users should understand the motif layer of KinaseCanvas as one particular, uniformly measured view of intrinsic specificity rather than as a complete model of substrate recognition.

5.3. *Orthogonality and information content of the evidence layers*

A reasonable concern about any resource that stacks filters is whether the layers are independent or largely re-encode the same signal. The analysis in Results shows that they are close to orthogonal, and the small redundancy that does exist, between curated status and functional score, is consistent with both depending on how thoroughly a site has been observed. This is expected from how the functional score is built. It is trained on 59 features spanning

583 mass-spectrometry evidence, phosphosite regulation, structural environment and
584 evolutionary conservation [5], and subcellular localization is not among them.
585 The co-localization filter therefore contributes essentially independent evidence
586 and does substantial work of its own, rather than rubber-stamping decisions
587 already made by the other layers.

588 5.4. *Biological insights from the integrated network*

589 The integrated network reveals several notable features of kinase-substrate
590 annotation. First, the distribution of novel substrates across kinases is highly
591 non-uniform (Figure 3(D)). Many kinases with already rich substrate annotations
592 gain additional predicted substrates but do not dominate the novel interaction
593 set, partly because of our promiscuity and functional filters. In contrast, a
594 large number of kinases with few or no curated substrates acquire predicted
595 substrates, often with strong motif support, high phosphosite functional scores
596 and clear co-localization with substrates (Figure 1(A); Figure 2(A); Figure 3(C)).
597 This pattern is particularly evident among dark kinases, for which more than
598 half of substrates are novel predictions in several kinase families (Figure 3(A)).
599 The fraction of KSI pairs involving dark kinases increases from 3.4% in curated
600 databases alone to 15.7% in the integrated network (Figure 3(B)), highlighting
601 the potential of motif-informed approaches to illuminate poorly characterised
602 parts of the kinome [14].

603 Second, our residue-level analyses show that tyrosine sites are both less frequent
604 and more strongly enriched for site-level support than serine or threonine sites:
605 36.1% of tyrosine sites in our resource are either already curated or carry
606 a functional score above the threshold, compared with 12.8% of serine and
607 15.0% of threonine sites (Figure 2(B)); the corresponding functional-score-only
608 figures, among sites carrying a score, are 26.5%, 7.7% and 8.6%. We report
609 this as a property of the assembled resource rather than as a claim about the
610 phosphoproteome, because several factors plausibly contribute to it and we
611 cannot separate them here. Much of the available phosphotyrosine data was
612 generated using anti-phosphotyrosine immunoaffinity enrichment [31], a targeted
613 design that both concentrates on sites of prior biological interest and differs
614 systematically from the untargeted enrichment used for serine and threonine
615 sites. Phosphotyrosine is also the smallest and best-studied of the three classes,
616 so its coverage is comparatively closer to saturation, and a substantial share of
617 well-characterised tyrosine sites are receptor tyrosine kinase autophosphorylation
618 sites whose documented function is to recruit SH2- and PTB-domain effectors
619 [32]. Any of these would raise the apparent functional-score enrichment of
620 tyrosine sites independently of underlying regulatory importance. Published
621 analyses likewise report that tyrosine and serine/threonine phosphorylation
622 differ in their regulatory character and in the depth to which each has been
623 sampled [3, 5], and the relative evolutionary conservation of phosphothreonine
624 and phosphotyrosine sites remains an area of active disagreement [4, 5]. We
625 therefore draw no evolutionary inference from this observation.

626 The value of layering orthogonal evidence is illustrated by RANGAP1 S442,
 627 an experimentally observed mitotic phosphosite that lies in a cyclin-dependent
 628 kinase (Cdk) consensus motif but has no assigned upstream kinase in the curated
 629 databases we integrated [33]. In the serine/threonine specificity atlas [15], five
 630 kinases score at or above the 90th percentile for this site: CDK2 (rank 1, 98.8th
 631 percentile), CDK3 (rank 2, 98.1st), CDK1 (rank 3, 94.8th), MARK3 (rank 4,
 632 90.6th) and GSK3B (rank 5, 90.2nd) – kinases with overlapping motif preferences
 633 that sequence information alone cannot separate. Localization does separate
 634 them: CDK1 and CDK3 share the cytosol with RANGAP1, whereas CDK2 and
 635 GSK3B are annotated only to compartments that RANGAP1 does not occupy
 636 and MARK3 has no HPA annotation. The site also carries a high functional
 637 score (0.92). The pipeline therefore retains CDK1 and CDK3, and its preference
 638 for CDK1 over CDK2 agrees with the original report, which proposed on the
 639 basis of the timing and location of RanGAP1’s mitotic phosphorylation that
 640 Cdk1, rather than the G1/S kinase Cdk2, is the likely *in vivo* kinase, although in
 641 that study recombinant Cdk2 phosphorylated T409 far more efficiently than S442
 642 *in vitro* [33]. Because CDK1 is already curated as a RANGAP1 kinase at the
 643 neighbouring site T409 (EPSD, iPTMnet and PhosphoSitePlus), this prediction is
 644 best read as a site-level extension of an established kinase-substrate relationship,
 645 of the kind the pair mode of the Explorer is designed to support. Localization
 646 is nevertheless a static, annotation-based filter, and here it removes a kinase
 647 that is capable of the reaction: recombinant cyclin A/Cdk2 also phosphorylates
 648 RanGAP1 *in vitro* [33].

649 Similarly, our resource illuminates potential regulatory roles for dark kinases.
 650 MAPK15 (ERK8) is classified as understudied by the IDG programme [14] and
 651 has few annotated substrates despite belonging to a well-characterised, proline-
 652 directed kinase family. Our pipeline nominates RING1, the E3 ubiquitin ligase
 653 subunit of the Polycomb repressive complex 1, as a candidate substrate at S38.
 654 The prediction satisfies every criterion in the pipeline and is internally coherent
 655 with what is known about the kinase: MAPK15 is the top-ranked kinase of the
 656 entire kinome for this site (rank 1, 99.0th percentile), the site carries a high
 657 functional score (0.89), MAPK15 and RING1 share two annotated compartments
 658 (cytosol and nucleoplasm), and the local sequence (DGTEIAV-S-PRSLHSE)
 659 is a canonical serine-proline motif of the kind MAPK-family kinases require.
 660 Six of the seven kinases that score at or above the 90th percentile for this site
 661 are retained (PINK1, ranked third, shares no compartment with RING1), and
 662 MAPK15 ranks first among them, well ahead of ERK1 (MAPK3; rank 2, 93.5th
 663 percentile). S38 is an experimentally observed phosphosite that currently has
 664 no assigned upstream kinase in the curated databases we integrated, which is
 665 precisely the gap this resource is intended to fill. We present it as a prioritised
 666 hypothesis for experimental testing rather than as an established mechanism.

667 Third, the family-wise patterns in Figure 1(E) and Figure 3(A) imply that
 668 certain kinase groups such as atypical and “other” kinases may be systematically
 669 under-annotated at the substrate level. For these groups, more than half of
 670 substrates in our resource are novel, motif-supported predictions (Figure 3(A)).

671 This suggests that a substantial fraction of phosphorylation events currently
672 attributed to generic “proline-directed” or “acidic” kinases could, in fact, be
673 mediated by less-characterised kinases with similar motifs but distinct biological
674 roles. Experimental testing of these candidates using targeted kinase perturbation
675 combined with phosphoproteomics or substrate-trapping mutants could reveal
676 new pathways and cross-talk among kinase families [18, 19, 34].

677 5.5. Limitations and caveats

678 Despite the stringent filtering, our predictions remain hypotheses and not
679 confirmed in vivo interactions. Several limitations merit consideration. First,
680 kinases with highly similar motifs (e.g., AKT vs. SGK, or different MAPKs) will
681 often share predicted substrates [15, 32, 35]. Sequence-based motif scoring alone
682 cannot always discriminate which kinase is responsible for a given phosphosite
683 in a particular cell type or condition. Our co-localization filter removes obvious
684 mismatches (e.g., a mostly cytosolic kinase and an exclusively nuclear substrate),
685 but does not capture dynamic relocalization, scaffold-mediated proximity or
686 compartment-specific isoforms. As a consequence, some true interactions may be
687 excluded, while some retained interactions may occur only in specific contexts
688 not captured by current localization data [22]. Users should therefore treat the
689 predicted kinase list for each phosphosite as a set of candidates to be prioritised
690 rather than a definitive assignment.

691 Second, we consider each phosphosite largely in isolation, whereas substrate
692 recognition in vivo depends on determinants that lie outside the +/-7 residue
693 window a position-specific matrix can see [32]. Kinases are recruited to substrates
694 through docking interactions distinct from the catalytic cleft, including the
695 D- and DEF-motifs used by MAP kinases and the SH2- and PTB-domain
696 interactions that bring receptor tyrosine kinase substrates to high effective local
697 concentration [32]. A peptide that scores poorly against a kinase’s linear motif
698 can therefore still be an efficient substrate once bound. Recognition is also
699 frequently hierarchical: casein kinase 1 requires a phosphorylated residue at
700 n-3, GSK3 at n+4, and casein kinase 2 can use either canonical or phospho-
701 determinants, so a site’s suitability depends on prior modification events that a
702 static motif score does not represent [32]. A site that matches a kinase motif
703 may equally remain unphosphorylated if it is buried in the folded protein or if
704 an upstream priming event is absent. Conversely, kinases can act on multiple
705 sites within the same substrate, and combinatorial patterns of phosphorylation
706 may be required for functional output. Our use of a phosphosite functional
707 score partially accounts for evolutionary conservation, structural features and
708 regulatory behaviour [5], but cannot fully model three-dimensional accessibility
709 or cooperative phosphorylation events. Taken together, these mechanisms bias
710 a top-percentile motif tier against precisely those kinases for which docking,
711 rather than the catalytic cleft, dominates substrate selection – which is one
712 reason we also release a continuous confidence score and a relaxed motif tier.
713 Incorporating structural information, as done in 3D-aware prediction tools, and

714 explicit modelling of docking and priming determinants would improve specificity
715 for kinases with overlapping linear motifs [32].

716 Third, every criterion is applied as a threshold, and thresholds are arbitrary at
717 their boundaries: a site scoring 0.4999 on the functional score is excluded while
718 one scoring 0.5001 is retained, and once retained the two are treated identically
719 even though a site scoring 0.99 is far better supported. The cut-offs are not
720 arbitrary in origin (0.5 is the value recommended by the authors of the functional
721 score [5], and the 90th motif percentile follows the specificity atlases [15, 16]),
722 and we mitigate their effect in two ways: the sensitivity analysis shows which
723 conclusions depend on the operating point, and the continuous confidence score
724 lets users rank rather than filter. A fully probabilistic treatment, in which each
725 layer contributes a calibrated likelihood, would be preferable, but it requires
726 labelled negative examples that do not exist at kinome scale; existing approaches
727 rely on curated positives with sampled decoys or on in vitro assay data [26, 27].

728 Fourth, our filters are spatial but not temporal. Requiring a kinase and substrate
729 to share a subcellular compartment is a necessary condition for a genuine
730 interaction, not a sufficient one: a real kinase-substrate pair should additionally
731 show coordinated behaviour over time, with substrate phosphorylation tracking
732 kinase activity across stimuli, inhibitor treatments or genetic perturbation.
733 Compendia of perturbation phosphoproteomics now make this tractable in
734 principle [36, 35], and adding a co-regulation layer – either as an additional
735 evidence component or as a filter in its own right – is the most obvious way to
736 improve the resource.

737 Fifth, coverage is systematically uneven with respect to cell-type-specific biol-
738 ogy: novel predictions reach 70.4% of substrates with low tissue specificity but
739 only 40.6% of tissue-enriched substrates. Three mechanisms contribute to this
740 result. The specificity atlases score only phosphosites already reported in phos-
741 phoproteomics experiments, a literature dominated by immortalized and cancer
742 cell lines, so a site never observed in those systems cannot enter the resource
743 however good its motif match [37]. Tissue-restricted proteins have sparser HPA
744 annotation, so co-localization can fail for missing-data rather than biological
745 reasons [22, 38]. Lastly, functional scores depend on depth of mass-spectrometry
746 sampling, which is likewise uneven. The practical consequence is that specialized
747 proteins that give differentiated cells their identity contribute the least beyond
748 existing curation. However, by the same logic, the resource is at its most useful
749 in settings where differentiated programs are lost: cancer being the clearest case,
750 and one where large phosphoproteomic cohorts already exist to test predictions
751 against. Deeper phosphoproteomics of primary tissues and better localization
752 coverage of tissue-restricted proteins would narrow this gap.

753 Finally, we focus on the presence of potential kinase-substrate relationships
754 rather than their quantitative strength or regulatory sign (activating vs. in-
755 hibitory). Integrating quantitative phosphoproteomic time series with prior KSI
756 networks has been shown to enable inference of causal directionality and sign
757 [21]. Extending our resource with such information is a promising direction but

758 beyond the scope of the current study.

759 5.6. Future directions

760 Several extensions could further enhance the resource. First, incremental
761 updates can incorporate new phosphosites, additional motif profiling experiments
762 and improved functional annotations. As kinase specificity profiling is completed
763 for remaining kinases and refined under different conditions [15, 16], our pipeline
764 can be rerun to refresh motif-based predictions and coverage. Second, dynamic
765 data could be used to refine or reweigh predicted interactions. Condition-
766 specific phosphoproteomics, kinase inhibition or knockdown experiments, and
767 proximity-labelling assays can provide direct evidence for or against particular
768 KSIs [5, 18, 34]. Such data could be integrated into probabilistic or machine-
769 learning models that jointly consider motif score, co-localization, co-expression,
770 network proximity and experimental perturbation response to yield calibrated
771 confidence estimates, as done in other kinase-substrate prediction frameworks
772 [5, 16, 18, 17, 11]. Our current filtering scheme could thus serve as a transparent
773 baseline on top of which more sophisticated predictive models are trained. Third,
774 tighter integration with existing resources would increase usability. For example,
775 predicted KSIs could be deposited into PhosphoSitePlus, SIGNOR or related
776 databases as “computationally inferred” interactions with explicit evidence codes
777 (motif-based, co-localization-supported, functional-site-supported) [23, 8, 11].
778 This would allow researchers querying a protein of interest to view predicted
779 kinases alongside curated ones. Similarly, linking our predictions with the
780 Dark Kinase Knowledgebase and other IDG tools [14] could help experimental
781 groups prioritize substrates for poorly studied kinases, accelerating efforts to
782 “de-orphanise” dark kinases.

783 Finally, there is an opportunity to exploit the resource in disease-focused studies.
784 Many of the substrates and sites in our network overlap with proteins mutated or
785 dysregulated in cancer and other diseases [23, 5, 21, 15, 17]. Combining our KSI
786 predictions with genomic or proteomic alterations from patient cohorts could
787 reveal candidate kinase-substrate axes driving pathology, and suggest biomarkers
788 or therapeutic targets whose activity might be inferred from phosphosite status.

789 6. Conclusion

790 In summary, we present a resource that substantially broadens the current
791 landscape of human kinase-substrate relationships by integrating empirical speci-
792 ficity atlases with curated phosphorylation databases, subcellular localization
793 and phosphosite functional scores. The integrated network increases the number
794 of candidate KSIs 3.6-fold, markedly increases coverage for dark kinases, and
795 enriches for sites with features indicative of regulatory function (Figure 1(A); Fig-
796 ure 2(A); Figure 3(C); Supp. Figure 1(A)). By design, the predicted interactions
797 are hypotheses that require experimental validation, but they provide a focused
798 and context-aware starting point for dissecting signalling pathways, particularly
799 in areas of the kinome that have received little prior attention. Bridging the gap

800 between large-scale phosphoproteomic data and specific kinase assignments is
801 essential for mechanistic understanding and for translating signalling knowledge
802 into therapeutic strategies [1, 7, 5, 28, 24]; our work represents a step toward
803 that goal.

804 7. Funding

805 This work was supported by the National Institutes of Health [R01 CA288145
806 to J.J.Y.; U01 CA274298 to J.J.Y. and X.L.P.; P50 CA257911 to J.J.Y. and
807 X.L.P.; U24 CA211000 to X.L.P. and J.J.Y.].

808 8. Conflict of Interest

809 The authors declare no competing interests.

810 9. Acknowledgements

811 We thank the contributors to the PhosphoSitePlus, EPSD, iPTMnet, and
812 Human Protein Atlas databases for providing the primary data that underlies
813 this integrated resource.

814 10. Data Availability Statement

815 The KinaseCanvas interactive resource is available at <https://cujoisa.github.io/KinaseCanvas/>, where the integrated network can be browsed
816 and the edge-level and site-level tables downloaded in full. Each released
817 kinase-substrate-site triplet carries the individual evidence values behind its
818 classification (motif percentile, the kinase’s rank for that site, promiscuity index,
819 phosphosite functional score, number of shared subcellular compartments and
820 the composite confidence score), so the inclusion criteria can be checked case by
821 case. All code used for data integration, filtering and analysis is available at
822 <https://github.com/cujoisa/KinaseCanvas>. Supplementary Tables S1-S4 (S1
823 being the complete unfiltered prediction matrix) and the relaxed motif tier
824 are deposited on Zenodo (<https://doi.org/10.5281/zenodo.22941318>); source
825 datasets are redistributed under the terms of their original licences.
826

827 11. Supplementary Data

828 Alt text: Three-panel figure showing phosphosite counts per database, the
829 percentage of functional sites across sources, and an UpSet plot visualizing source
830 overlaps.

Unfiltered specificity atlas. Motif ranks, percentiles, and promiscuity indices for the human kinome specificity atlases. Because of its size (27.8 million rows; 3.6 GB), this table is available from Zenodo (<https://doi.org/10.5281/zenodo.22941318>) as [results/unfiltered_cantley_dataset.csv](#).

Supp. Table 1

Subcellular localization mapping. Protein compartmentalization data derived from the Human Protein Atlas (HPA). Provided as [Supplementary_Table_S2.csv](#).

Supp. Table 2

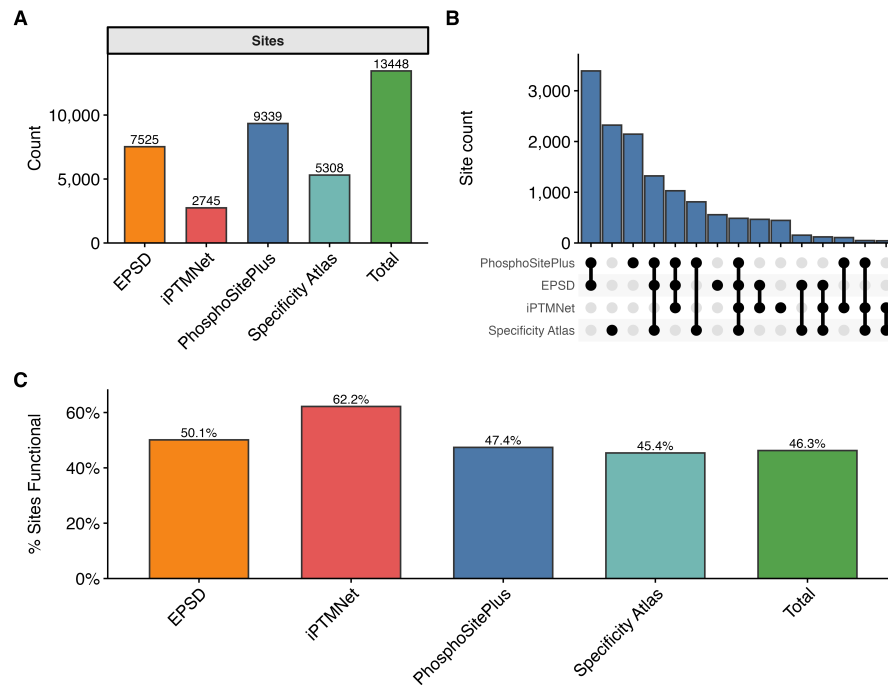
Phosphosite functional scores. Pre-computed functional scores for each reported phosphosite in the human phosphoproteome. Provided as [Supplementary_Table_S3.csv](#).

Supp. Table 3

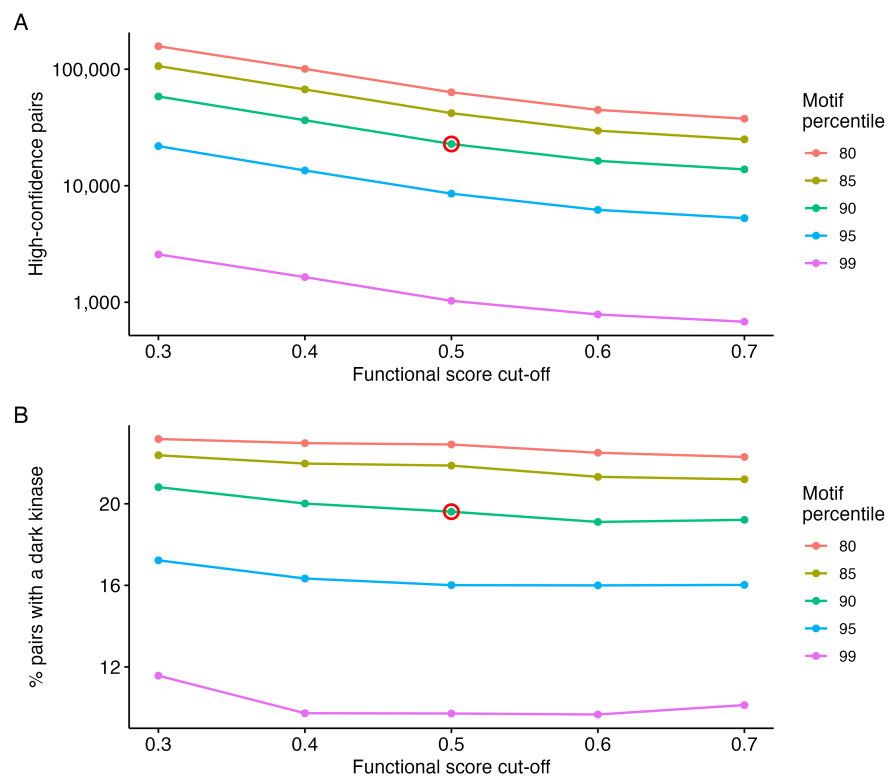
Substrate counts at each stage of the filtering summarised in Figure 1C. Each substrate is counted once per stage. A substrate enters the diagram if at least one of its sites ranks at or above the 90th percentile for a kinase with below-median promiscuity; it is “known” if it has at least one curated interaction with any kinase and “novel” otherwise. Site-level support means that at least one such site is curated or has a functional score ≥ 0.5 . A substrate is retained (“co-localized”) when at least one of its motif-passing kinase-site pairs has a supported site and a kinase that shares an HPA compartment with the substrate, so the final stage is the set of substrates contributed by the specificity atlas to the integrated resource. Substrates without site-level support are not carried into the last two stages. Provided as [Supplementary_Table_S4.csv](#).

Stage	Category	Substrates
Specificity Atlas	Known substrate	2,697
Specificity Atlas	Novel substrate	6,078
Site support	Curated + functional score	800
Site support	Curated site	929
Site support	Functional score only	1,578
Site support	No site-level support	5,468
Co-localized?	Co-localized	2,471
Co-localized?	No co-localization	836
Filtered Result	Known substrate	1,602
Filtered Result	Novel substrate	869

Supp. Table 4



Supp. Figure 1: **Comprehensive landscape and functional density of integrated phosphosites.** A) Distribution of unique phosphosite counts across individual database sources (PhosphoSitePlus, EPSP, iPTMnet) and the predicted specificity atlas, alongside the non-redundant integrated total. B) UpSet plot showing the intersection of unique phosphosites across the four primary sources. While many sites are unique to specific repositories, KinaseCanvas provides a unified framework that highlights the substantial expansion provided by integrating motif-based predictions. C) Percentage of phosphosites within each source designated as 'likely functional' (Ochoa functional score ≥ 0.5), demonstrating high functional density across both curated and predicted datasets.



Supp. Figure 2: **Threshold sensitivity of the high-confidence interaction set.** A) Number of retained kinase-substrate pairs as a function of the phosphosite functional-score cut-off, at motif percentile thresholds from the 80th to the 99th. Note the logarithmic y-axis. B) Percentage of retained pairs involving a kinase classified as understudied (dark) by the IDG programme, over the same grid. The operating point used throughout the manuscript (90th motif percentile, functional score 0.5) is circled in red in both panels. The absolute size of the resource depends strongly on the thresholds chosen, whereas the dark-kinase share stays between 9.7% and 23.2% across the whole grid, well above the 3.4% of curated pairs.

References

- [1] P. V. Hornbeck, J. M. Kornhauser, V. Latham, B. Murray, V. Nandhikonda, A. Nord, E. Skrzypek, T. Wheeler, B. Zhang, F. Gnad, 15 years of PhosphoSitePlus: integrating post-translationally modified sites, disease variants and isoforms, *Nucleic Acids Research* 47 (D1) (2019) D433–D441. doi:10.1093/nar/gky1159.
- [2] P. Vlastaridis, P. Kyriakidou, A. Chaliotis, Y. Van de Peer, S. G. Oliver, G. D. Amoutzias, Estimating the total number of phosphoproteins and phosphorylation sites in eukaryotic proteomes, *GigaScience* 6 (2) (2017) 1–11. doi:10.1093/gigascience/giw015.
- [3] K. Sharma, R. C. J. D’Souza, S. Tyanova, C. Schaab, J. R. Wiśniewski, J. Cox, M. Mann, Ultradeep human phosphoproteome reveals a distinct regulatory nature of Tyr and Ser/Thr-based signaling, *Cell Reports* 8 (5) (2014) 1583–1594. doi:10.1016/j.celrep.2014.07.036.
- [4] C. R. Landry, E. D. Levy, S. W. Michnick, Weak functional constraints on phosphoproteomes, *Trends in Genetics* 25 (5) (2009) 193–197. doi:10.1016/j.tig.2009.03.003.
- [5] D. Ochoa, A. F. Jarnuczak, C. Viéitez, M. Gehre, et al., The functional landscape of the human phosphoproteome, *Nature Biotechnology* 38 (2020) 365–373. doi:10.1038/s41587-019-0344-3.
- [6] A. Kalyuzhnyy, P. A. Eyers, C. E. Eyers, E. Bowler-Barnett, M. J. Martin, Z. Sun, E. W. Deutsch, A. R. Jones, Profiling the human phosphoproteome to estimate the true extent of protein phosphorylation, *Journal of Proteome Research* 21 (6) (2022) 1510–1524. doi:10.1021/acs.jproteome.2c00131.
- [7] M. Chen, Y. Gou, M. Lei, L. Xiao, M. Zhao, X. Huang, D. Liu, Z. Feng, D. Peng, Y. Xue, EPSD 2.0: An updated database of protein phosphorylation sites across eukaryotic species, *Genomics, Proteomics & Bioinformatics* 23 (3) (2025) qzaf057. doi:10.1093/gpbjnl/qzaf057.
- [8] H. Huang, C. N. Arighi, K. E. Ross, J. Ren, G. Li, S.-C. Chen, Q. Wang, J. Cowart, K. Vijay-Shanker, C. H. Wu, iPTMnet: an integrated resource for protein post-translational modification network discovery, *Nucleic Acids Research* 46 (D1) (2018) D542–D550. doi:10.1093/nar/gkx1104.
- [9] H. Dinkel, C. Chica, A. Via, C. M. Gould, L. J. Jensen, T. J. Gibson, F. Diella, Phospho.ELM: a database of phosphorylation sites—update 2011, *Nucleic Acids Research* 39 (Database issue) (2011) D261–D267. doi:10.1093/nar/gkq1104.
- [10] J. Hu, H.-S. Rho, R. H. Newman, J. Zhang, H. Zhu, J. Qian, Phospho-Networks: a database for human phosphorylation networks, *Bioinformatics* 30 (1) (2014) 141–142. doi:10.1093/bioinformatics/btt627.

- [11] L. Licata, P. Lo Surdo, M. Iannuccelli, A. Palma, E. Micarelli, L. Perfetto, D. Peluso, A. Calderone, L. Castagnoli, G. Cesareni, SIGNOR 2.0, the SIGnaling network open resource 2.0: 2019 update, *Nucleic Acids Research* 48 (D1) (2020) D504–D510. doi:10.1093/nar/gkz949.
- [12] R. H. Newman, J. Hu, H.-S. Rho, Z. Xie, C. Woodard, J. Neiswinger, C. Cooper, M. Shirley, H. M. Clark, S. Hu, et al., Construction of human activity-based phosphorylation networks, *Molecular Systems Biology* 9 (2013) 655. doi:10.1038/msb.2013.12.
- [13] J. Safaei, J. Mañuch, A. Gupta, L. Stacho, S. Pelech, Prediction of 492 human protein kinase substrate specificities, *Proteome Science* 9 (Suppl 1) (2011) S6. doi:10.1186/1477-5956-9-S1-S6.
- [14] M. E. Berginski, N. Moret, C. Liu, D. Goldfarb, P. K. Sorger, S. M. Gomez, The dark kinase knowledgebase: an online compendium of knowledge and experimental results of understudied kinases, *Nucleic Acids Research* 49 (D1) (2021) D529–D535. doi:10.1093/nar/gkaa853.
- [15] J. L. Johnson, T. M. Yaron, E. M. Huntsman, A. Kerelsky, J. Song, A. Regev, T.-Y. Lin, K. Liberatore, D. M. Cizin, B. M. Cohen, N. Vasan, Y. Ma, K. Krismer, J. T. Robles, B. van de Kooij, A. E. van Vlimmeren, N. André-Busch, N. F. Käufer, M. V. Dorovkov, A. G. Ryazanov, Y. Takagi, E. R. Kastenhuber, M. D. Goncalves, B. D. Hopkins, O. Elemento, D. J. Taatjes, A. Maucuer, A. Yamashita, A. Degterev, M. Uduman, J. Lu, S. D. Landry, B. Zhang, I. Cossentino, R. Linding, J. Blenis, P. V. Hornbeck, B. E. Turk, M. B. Yaffe, L. C. Cantley, An atlas of substrate specificities for the human serine/threonine kinome, *Nature* 613 (7945) (2023) 759–766. doi:10.1038/s41586-022-05575-3.
- [16] T. M. Yaron-Barir, B. A. Joughin, E. M. Huntsman, A. Kerelsky, D. M. Cizin, B. M. Cohen, A. Regev, J. Song, N. Vasan, T.-Y. Lin, J. M. Orozco, C. Schoenherr, C. Sagum, M. T. Bedford, R. M. Wynn, S.-C. Tso, D. T. Chuang, L. Li, S. S.-C. Li, P. Creixell, K. Krismer, M. Takegami, H. Lee, B. Zhang, J. Lu, I. Cossentino, S. D. Landry, M. Uduman, J. Blenis, O. Elemento, M. C. Frame, P. V. Hornbeck, L. C. Cantley, B. E. Turk, M. B. Yaffe, J. L. Johnson, The intrinsic substrate specificity of the human tyrosine kinome, *Nature* 629 (8014) (2024) 1174–1181. doi:10.1038/s41586-024-07407-y.
- [17] J. C. Obenauer, L. C. Cantley, M. B. Yaffe, Scansite 2.0: Proteome-wide prediction of cell signaling interactions using short sequence motifs, *Nucleic Acids Research* 31 (13) (2003) 3635–3641. doi:10.1093/nar/gkg584.
- [18] M. L. Miller, L. J. Jensen, F. Diella, C. Jørgensen, M. Tinti, L. Li, M. Hsiung, S. A. Parker, J. Bordeaux, T. Sicheritz-Ponten, M. Olhovsky, A. Pasculescu, J. Alexander, S. Knapp, N. Blom, P. Bork, S. Li, G. Cesareni, T. Pawson, B. E. Turk, M. B. Yaffe, S. Brunak, R. Linding, Linear motif atlas for

- 911 phosphorylation-dependent signaling, *Science Signaling* 1 (35) (2008) ra2.
912 [doi:10.1126/scisignal.1159433](https://doi.org/10.1126/scisignal.1159433).
- 913 [19] R. Linding, L. J. Jensen, G. J. Ostheimer, et al., Systematic discovery of
914 in vivo phosphorylation networks, *Cell* 129 (7) (2007) 1415–1426. [doi:](https://doi.org/10.1016/j.cell.2007.05.052)
915 [10.1016/j.cell.2007.05.052](https://doi.org/10.1016/j.cell.2007.05.052).
- 916 [20] L. Higgins, H. Gerdes, P. R. Cutillas, Principles of phosphoproteomics and
917 applications in cancer research, *Biochemical Journal* 480 (6) (2023) 403–420.
918 [doi:10.1042/BCJ20220220](https://doi.org/10.1042/BCJ20220220).
- 919 [21] B. M. Invergo, B. Petursson, N. Akhtar, D. Bradley, G. Giudice, M. Hijazi,
920 P. Cutillas, E. Petsalaki, P. Beltrao, Prediction of signed protein kinase
921 regulatory circuits, *Cell Systems* 10 (5) (2020) 384–396.e9. [doi:10.1016/](https://doi.org/10.1016/j.cels.2020.04.005)
922 [j.cels.2020.04.005](https://doi.org/10.1016/j.cels.2020.04.005).
- 923 [22] P. J. Thul, L. Åkesson, M. Wiking, D. Mahdessian, A. Geladaki, H. Ait Blal,
924 T. Alm, A. Asplund, L. Björk, L. M. Breckels, A. Bäckström, F. Danielsson,
925 L. Fagerberg, J. Fall, L. Gatto, C. Gnann, S. Hober, M. Hjelmare,
926 F. Johansson, S. Lee, C. Lindskog, J. Mulder, C. M. Mulvey, P. Nilsson,
927 P. Oksvold, J. Rockberg, R. Schutten, J. M. Schwenk, Å. Sivertsson,
928 E. Sjöstedt, M. Skogs, C. Stadler, D. P. Sullivan, H. Tegel, C. Winsnes,
929 C. Zhang, M. Zwahlen, A. Mardinoglu, F. Pontén, K. von Feilitzen, K. S.
930 Lilley, M. Uhlén, E. Lundberg, A subcellular map of the human proteome,
931 *Science* 356 (6340) (2017) eaal3321. [doi:10.1126/science.aal3321](https://doi.org/10.1126/science.aal3321).
- 932 [23] P. V. Hornbeck, B. Zhang, B. Murray, J. M. Kornhauser, V. Latham,
933 E. Skrzypek, PhosphoSitePlus, 2014: mutations, PTMs and recalibrations,
934 *Nucleic Acids Research* 43 (Database issue) (2015) D512–D520. [doi:](https://doi.org/10.1093/nar/gku1267)
935 [10.1093/nar/gku1267](https://doi.org/10.1093/nar/gku1267).
- 936 [24] J. A. P. Sekar, Y. C. Li, A. Schlessinger, G. Pandey, A web portal for ex-
937 ploring kinase-substrate interactions, *npj Systems Biology and Applications*
938 10 (1) (2024) 113. [doi:10.1038/s41540-024-00442-5](https://doi.org/10.1038/s41540-024-00442-5).
- 939 [25] D. Wang, D. Liu, J. Yuchi, F. He, Y. Jiang, S. Cai, J. Li, D. Xu, MusiteDeep:
940 a deep-learning based webserver for protein post-translational modification
941 site prediction and visualization, *Nucleic Acids Research* 48 (W1) (2020)
942 W140–W146. [doi:10.1093/nar/gkaa275](https://doi.org/10.1093/nar/gkaa275).
- 943 [26] B. M. Invergo, Accurate, high-coverage assignment of in vivo protein kinases
944 to phosphosites from in vitro phosphoproteomic specificity data, *PLOS*
945 *Computational Biology* 18 (5) (2022) e1010110. [doi:10.1371/journal.](https://doi.org/10.1371/journal.pcbi.1010110)
946 [pcbi.1010110](https://doi.org/10.1371/journal.pcbi.1010110).
- 947 [27] H. Horn, E. M. Schoof, J. Kim, X. Robin, M. L. Miller, F. Diella, A. Palma,
948 G. Cesareni, L. J. Jensen, R. Linding, KinomeXplorer: an integrated
949 platform for kinome biology studies, *Nature Methods* 11 (6) (2014) 603–604.
950 [doi:10.1038/nmeth.2968](https://doi.org/10.1038/nmeth.2968).

- [28] C. Wang, H. Xu, S. Lin, W. Deng, J. Zhou, Y. Zhang, Y. Shi, D. Peng, Y. Xue, GPS 5.0: An update on the prediction of kinase-specific phosphorylation sites in proteins, *Genomics, Proteomics & Bioinformatics* 18 (1) (2020) 72–80. doi:10.1016/j.gpb.2020.01.001.
- [29] N. Sugiyama, H. Imamura, Y. Ishihama, Large-scale discovery of substrates of the human kinome, *Scientific Reports* 9 (1) (2019) 10503. doi:10.1038/s41598-019-46385-4.
- [30] B. G. Poll, K. T. Leo, V. Deshpande, N. Jayatissa, T. Pisitkun, E. Park, C.-R. Yang, V. Raghuram, M. A. Knepper, A resource database for protein kinase substrate sequence-preference motifs based on large-scale mass spectrometry data, *Cell Communication and Signaling* 22 (1) (2024) 137. doi:10.1186/s12964-023-01436-2.
- [31] J. Rush, A. Moritz, K. A. Lee, A. Guo, V. L. Goss, E. J. Spek, H. Zhang, X.-M. Zha, R. D. Polakiewicz, M. J. Comb, Immunoaffinity profiling of tyrosine phosphorylation in cancer cells, *Nature Biotechnology* 23 (1) (2005) 94–101. doi:10.1038/nbt1046.
- [32] C. J. Miller, B. E. Turk, Homing in: mechanisms of substrate targeting by protein kinases, *Trends in Biochemical Sciences* 43 (5) (2018) 380–394. doi:10.1016/j.tibs.2018.02.009.
- [33] S. Swaminathan, F. Kiendl, R. Körner, R. Lupetti, L. Hengst, F. Melchior, RanGAP1*SUMO1 is phosphorylated at the onset of mitosis and remains associated with RanBP2 upon NPC disassembly, *The Journal of Cell Biology* 164 (7) (2004) 965–971. doi:10.1083/jcb.200309126.
- [34] J. S. Gerritsen, F. M. White, Phosphoproteomics: a valuable tool for uncovering molecular signaling in cancer cells, *Expert Review of Proteomics* 18 (8) (2021) 661–674. doi:10.1080/14789450.2021.1976152.
- [35] S. Müller-Dott, E. J. Jaehnig, K. P. Munchic, W. Jiang, T. M. Yaron-Barir, S. R. Savage, M. Garrido-Rodriguez, J. L. Johnson, A. Lussana, E. Petsalaki, J. T. Lei, A. Dugourd, K. Krug, L. C. Cantley, D. R. Mani, B. Zhang, J. Saez-Rodriguez, Comprehensive evaluation of phosphoproteomic-based kinase activity inference, *Nature Communications* 16 (1) (2025) 4771. doi:10.1038/s41467-025-59779-y.
- [36] D. Ochoa, M. Jonikas, R. T. Lawrence, B. El Debs, J. Selkrig, A. Tympas, J. Villén, S. D. M. Santos, P. Beltrao, An atlas of human kinase regulation, *Molecular Systems Biology* 12 (12) (2016) 888. doi:10.15252/msb.20167295.
- [37] E. L. Huttlin, M. P. Jedrychowski, J. E. Elias, T. Goswami, R. Rad, S. A. Beausoleil, J. Villén, W. Haas, M. E. Sowa, S. P. Gygi, A tissue-specific atlas of mouse protein phosphorylation and expression, *Cell* 143 (7) (2010) 1174–1189. doi:10.1016/j.cell.2010.12.001.

991 [38] M. Uhlén, L. Fagerberg, B. M. Hallström, C. Lindskog, P. Oksvold,
992 A. Mardinoglu, Å. Sivertsson, C. Kampf, E. Sjöstedt, A. Asplund, I. Ols-
993 son, K. Edlund, E. Lundberg, S. Navani, C. A.-K. Szigyanto, J. Odeberg,
994 D. Djureinovic, J. O. Takanen, S. Hober, T. Alm, P.-H. Edqvist, H. Berling,
995 H. Tegel, J. Mulder, J. Rockberg, P. Nilsson, J. M. Schwenk, M. Hamsten,
996 K. von Feilitzen, M. Forsberg, L. Persson, F. Johansson, M. Zwahlen, G. von
997 Heijne, J. Nielsen, F. Pontén, Tissue-based map of the human proteome,
998 Science 347 (6220) (2015) 1260419. [doi:10.1126/science.1260419](https://doi.org/10.1126/science.1260419).